

SC42035 Integration Project Systems and Control

by

Group 10

Xu Yan - 6367038

Aerospace Engineering

&

Diaa Abbasi - 5490626

Aerospace Engineering

The Github repository link is available at:

`git@github.com:DIAAAAAAAAAA/Helicopter_sim.git.`

Responsible Instructor: Dr. L. Laurenti

Instructors: Ir. S.C. Bregman & Ing. M.A. Zagorowska

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Delft University of Technology

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Nomenclature

Symbol	Description	Unit
I_{yy}	Pitch-axis moment of inertia	$kg \cdot m^2$
m_{heli}	Helicopter body mass	kg
m_{cw}	Counterweight mass	kg
L_{heli}	Distance from pivot to helicopter center of mass	m
L_{cw}	Distance from pivot to counterweight center of mass	m
L_m	Distance from pivot to motor thrust line	m
θ	Pitch angle	rad
$\dot{\theta}$	Pitch angular velocity	rad/s
$\ddot{\theta}$	Pitch angular acceleration	rad/s^2
T	Rotor thrust force	N
T_{bias}	Constant bias torque	$N \cdot m$
C_{pitch}	Pitch damping coefficient	$N \cdot m \cdot s/rad$
u	Motor input voltage	V
I	Armature current	A
ω	Rotor angular velocity	rad/s
R	Armature resistance	Ω
L	Armature inductance	H
K_m	Motor constant	$V \cdot s/rad$
J	Rotor inertia	$kg \cdot m^2$
c	Rotor drag coefficient	$N \cdot m \cdot s^2/rad^2$
C_T	Thrust coefficient	$N \cdot s^2/rad^2$
g	Gravitational acceleration	m/s^2
x	State vector	—
y	Measured output	rad
A, B, C, D	Linearized state-space matrices	—
K_x	State-feedback gain	—
K_i	Integral gain	—
K_f	Observer gain matrix	—
Q_{aug}	LQI state weighting matrix	—
R	LQI input weighting scalar	—
W	LQE process noise covariance matrix	—
V	LQE measurement noise covariance	—
T_s	Sampling time	s
N_p	MPC prediction horizon	samples
N_c	MPC control horizon	samples
Q	MPC output tracking weight	—
R	MPC input-rate weight	—
F_s	Sampling frequency	Hz
f_{start}	Starting frequency of Chirp input	Hz
f_{end}	Final frequency of Chirp input	Hz
T_{period}	Duration of one Chirp period	s
$V_{p,hover}$	Hover-offset voltage for Chirp excitation	V

1 Introduction

This report investigates the modeling, identification, and control of a 1-DOF laboratory helicopter with motion restricted to the pitch axis. The setup could be seen in Figure 1. The overall objective is to obtain a physically meaningful dynamic model of the system and to use this model as a basis for the design of linear controllers capable of accurate pitch-angle tracking around the hovering operating point.

The work was carried out in three main stages. First, a nonlinear dynamic model was derived by combining the mechanical equation of motion of the pitch axis with the electrical and rotational dynamics of the DC motor-driven rotor. Second, the unknown model parameters were identified experimentally using a two-stage grey-box identification procedure. To reduce the number of free parameters and improve parameter identifiability, the mechanical subsystem was first estimated from a free-swinging pendulum experiment under zero motor input. After this, a Chirp excitation signal was applied to identify the remaining electrical and thrust-related parameters under active operation.

Finally, the identified nonlinear model was linearized around the hovering equilibrium point and used for the design of two linear controllers: a Linear Quadratic Gaussian (LQG) Tracker and a Model Predictive Controller (MPC). The LQG approach combines optimal state feedback, integral action, and state estimation, whereas the MPC approach uses online optimization over a finite prediction horizon. The performance of both controllers was assessed in simulation in order to evaluate their suitability for pitch-angle regulation of the helicopter.

The remainder of this report is organized as follows. Chapter 2 presents the derivation of the nonlinear system model. Chapter 3 describes the parameter estimation procedure and the corresponding validation results. In chapter 4, the two controllers are designed and their performance is evaluated and compared.



Figure 1: 1-DOF laboratory helicopter setup

2 Modeling

This section develops the dynamic model of a 1-DOF helicopter, focusing on the motion characteristics of the pitch angle (θ). The system utilizes a DC motor-driven rotor to generate thrust, which acts against gravitational restoring torque and mechanical damping, thereby enabling controlled pitch-angle motion. The helicopter free-body diagram is shown in Figure 2.

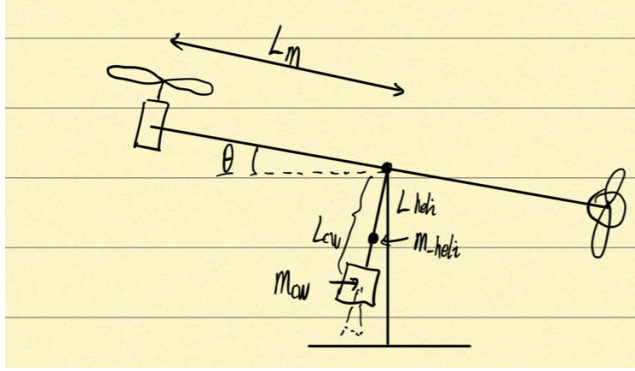


Figure 2: Free-body diagram of the 1-DOF helicopter pitch axis.

2.1 Modeling Assumptions

To obtain a tractable model suitable for identification and controller synthesis, several simplifying assumptions are introduced. First, the helicopter is treated as a rigid body whose motion is restricted to the pitch axis only; therefore, cross-coupling effects with yaw and roll are neglected. Second, the aerodynamic thrust generated by the rotor is approximated by a quadratic relation, $T = C_T \omega^2$, which captures the dominant dependence of lift on rotor speed in the operating region of interest. Third, the rotor and drivetrain losses are lumped into the nonlinear drag term $c\omega^2$. Finally, unmodeled effects such as blade flapping, structural flexibility, actuator saturation, and high-frequency aerodynamic disturbances are not explicitly included in the state-space equations. These assumptions are justified by the control objective, which focuses on accurate pitch-angle regulation around the hovering condition.

2.2 Mechanical system EOM

The generalized coordinate for the system is chosen as the pitch angle θ . Under the rigid-body assumption, the kinetic energy T_{sys} and potential energy V_{sys} are defined as follows:

$$T_{sys} = \frac{1}{2} I_{yy} \dot{\theta}^2 \quad (1)$$

$$V_{sys} = (m_{heli} L_{heli} + m_{cw} L_{cw}) g (1 - \cos(\theta)) \quad (2)$$

where I_{yy} denotes the pitch-axis moment of inertia, m_{heli} and m_{cw} are the masses of the helicopter body and counterweight, respectively. L_{heli} and L_{cw} are the corresponding distances from the pivot to their centers of mass.

The Lagrangian of the system is defined as

$$L = T_{sys} - V_{sys} \quad (3)$$

Applying the Euler-Lagrange equation yields

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q_{nc} \quad (4)$$

where Q_{nc} represents the non-conservative generalized torque acting on the pitch axis. In the present case, this torque consists of the rotor thrust moment, viscous pitch damping, and a constant bias torque:

$$Q_{nc} = T \cdot L_m - C_{pitch} \dot{\theta} + T_{bias} \quad (5)$$

Substituting these expressions into the Euler-Lagrange equation and rearranging gives the nonlinear pitch-axis equation of motion:

$$I_{yy} \ddot{\theta} = T \cdot L_m - (m_{heli} L_{heli} + m_{cw} L_{cw}) g \sin(\theta) - C_{pitch} \dot{\theta} + T_{bias} \quad (6)$$

This equation shows that the pitch acceleration is determined by a balance between the rotor-generated actuation torque, the gravitational restoring torque, the dissipative damping torque, and the constant torque bias.

2.3 Electrical Subsystem and State-Space Model

The thrust force is generated by a DC motor-driven rotor. The corresponding electrical circuit is shown in Figure 3.

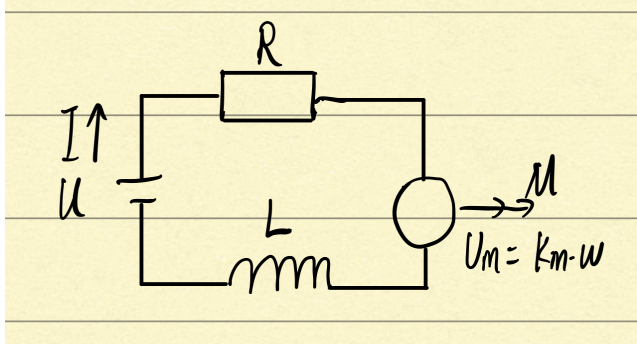


Figure 3: Motor circuit

According to Figure 3. The electrical characteristics and dynamics of the DC motor are described by the following equations:

$$\dot{I} = \frac{u - RI - K_m \omega}{L} \quad (7)$$

$$\dot{\omega} = \frac{K_m I - c \omega^2}{J} \quad (8)$$

where u is the applied motor voltage, I is the armature current, ω is the rotor angular velocity, R is the armature resistance, L is the armature inductance, K_m is the motor constant, J is the rotor inertia, and c is a lumped aerodynamic drag coefficient. The thrust produced by the rotor is modeled by the quadratic relation

$$T = C_T \omega^2 \quad (9)$$

where C_T is the thrust coefficient. Furthermore, by defining the state vector as

$$x = [x_1 \ x_2 \ x_3 \ x_4]^T = [\theta \ \dot{\theta} \ I \ \omega]^T \quad (10)$$

the complete nonlinear state-space model becomes

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \frac{1}{I_{yy}} [(C_T x_4^2) L_m - (m_{heli} L_{heli} + m_{cw} L_{cw}) g \sin(x_1) - C_{pitch} x_2 + T_{bias}] \\ \dot{x}_3 = \frac{1}{L} (u - R x_3 - K_m x_4) \\ \dot{x}_4 = \frac{1}{J} (K_m x_3 - c x_4^2) \end{cases} \quad (11)$$

with measured output

$$y = x_1 = \theta \quad (12)$$

3 System identification

The complete nonlinear model contains both mechanical and actuator-related parameters. In order to improve parameter identifiability and reduce the complexity of the optimization problem, the identification procedure was divided into two stages. First, the dominant mechanical parameters were identified from a free-swinging pendulum experiment with zero motor input. Subsequently, once the mechanical dynamics were characterized, the electrical and thrust-related parameters were estimated from an actively excited experiment using a Chirp input signal.

3.1 Identification of Mechanical Parameters

The identification of the mechanical parameters, specifically the moment of inertia (I_{yy}), helicopter body mass (m_{heli}), pitch damping coefficient (C_{pitch}), and constant bias torque (T_{bias}), was conducted using a free-swinging pendulum experiment. By setting the input voltage to zero, the structural mechanical dynamics is isolated from the electrical and thrust subsystems.

This staged identification strategy is adopted to improve parameter identifiability. If all mechanical, electrical, and thrust-related parameters were estimated simultaneously, strong correlations between parameter effects could lead to poor convergence or non-physical estimates. By first isolating the free mechanical motion of the system, the dominant gravitational and damping dynamics can be identified with high confidence. The resulting estimates then provide a physically meaningful baseline for the second-stage identification under active motor excitation.

3.2 Experimental Data and Model Configuration

The data used for the mechanical parameter estimation was obtained during a period of free oscillation ($T = 28.81$ s to 40.35 s). During this interval, the system behaves as a damped nonlinear pendulum subject to a constant torque bias. In order to estimate the relevant parameters within a grey-box framework, the nonlinear state-space model was used with the electrical states initialized at zero and effectively suppressed, such that the model reduces to the following pitch dynamics:

$$I_{yy}\ddot{\theta} = T_{bias} - (m_{heli}L_{heli} + m_{cw}L_{cw})g\sin(\theta) - C_{pitch}\dot{\theta} \quad (13)$$

The estimation was carried out using the `nlgreyest` routine with the `lsqnonlin` search algorithm. Physical lower and upper bounds were imposed on selected parameters, particularly I_{yy} and m_{heli} , in order to maintain physically meaningful estimates throughout the optimization process.

3.3 Identification Results and Model Validation

The estimation process converged in 12 iterations. As shown in the identification progress results, the cost function reached a steady minimum of 1.25624×10^{-6} .

The model achieves an excellent fit of **97.88%** to the measured data. Figure 4 demonstrates the high fidelity between the experimental free-swing response and the simulated output of the identified grey-box model. The high fit percentage confirms that the derived nonlinear mechanical model captures the dominant passive pitch dynamics of the helicopter. In particular, the gravitational restoring torque, the damping-induced energy dissipation, and the constant trimming effect caused by T_{bias} are all represented with high fidelity.

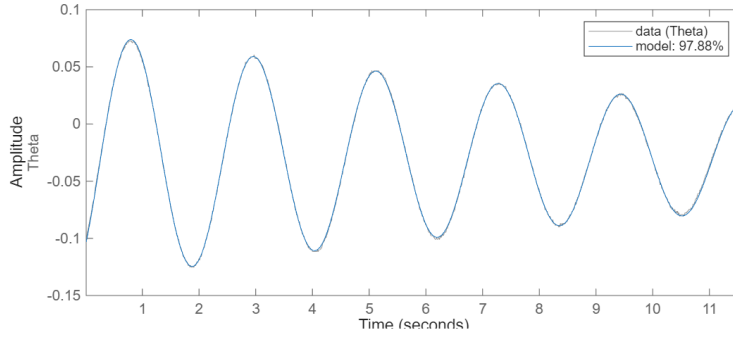


Figure 4: Fit of the identified nonlinear grey-box model to the experimental pendulum data (Fit: 97.88%).

3.4 Identified Parameter Values

Table 2 summarizes the identified mechanical parameters. These values form the basis for the second stage of the identification procedure, in which the actuator and thrust dynamics are estimated while the mechanical parameters are fixed.

Table 2: Identified Mechanical Parameters

Parameter	Estimated Value	Unit
I_{yy}	0.0399	$kg \cdot m^2$
m_{heli}	0.7952	kg
C_{pitch}	0.0039	$N \cdot m \cdot s/rad$
T_{bias}	-0.0092	$N \cdot m$

These identified values are physically meaningful. The parameter I_{yy} describes the resistance of the helicopter structure to pitch acceleration, while C_{pitch} quantifies the dissipative effects present in the rotational motion. The non-zero bias torque T_{bias} indicates the presence of a constant asymmetry in the setup, such

as a slight imbalance in mass distribution or an unmodeled offset torque at the pivot.

3.5 Identification of Electrical and Thrust Dynamics

Following the characterization of mechanical parameters, the electrical and thrust subsystems were identified using a Frequency Sweep excitation. In this second stage, the previously identified mechanical parameters were fixed. A Chirp signal was applied to the pitch motor to ensure sufficient persistent excitation across the system’s operational frequency range (0.1 Hz to 2.5 Hz), allowing for the identification of motor electrical constants (R, L, K_m, J) and the rotor thrust coefficient (C_T). This reduced the number of free parameters in the optimization and improved the identifiability of the actuator dynamics.

3.6 Excitation Signal Design

The use of a frequency sweep (Chirp) signal for system identification is motivated by its ability to provide persistent excitation across a broad frequency spectrum in a single experiment. Unlike impulsive or step inputs, which are often limited to specific frequency bands or require high magnitudes that risk actuator saturation, the Chirp signal systematically traverses the entire frequency range of interest in a single experiment. This allows for the simultaneous excitation of both low-frequency mechanical modes—such as pendulum-like oscillations—and high-frequency electrical dynamics, including motor electromagnetic responses. Consequently, this method enables the effective decoupling and identification of distinct physical parameters, such as inertia, damping, and electrical constants. In essence, the Chirp signal acts as a comprehensive ‘spectral scan’ of the system, ensuring that the resulting model captures an accurate and robust mapping of the system’s dynamics across all operational conditions.

To achieve comprehensive model identification, the input voltage V_p was designed as a Chirp signal superimposed on a hover-offset voltage ($V_{p,hover} = 0.15$ V):

$$V_p(t) = V_{p,hover} + A_p \cdot \text{chirp}(t, f_{start}, T_{period}, f_{end}) \quad (14)$$

This signal ensures that the system dynamics are captured under active control, with a sampling frequency of $F_s = 100$ Hz to make sure the induced time delay is small enough.

The hover-offset voltage serves two important purposes. First, it maintains the rotor within a positive-speed operating region, ensuring continuous thrust generation and avoiding counter-rotation of the motor. Second, it excites the system around the same neighborhood in which the final controller is intended to operate, thereby improving the relevance of the identified model for subsequent linearization and regulation. In this sense, the Chirp experiment is not only a

system identification procedure, but also a means of characterizing the plant around its practical working point.

3.7 Data Processing and Validation

The measured response was divided into repeated Chirp periods in order to obtain a consistent estimation dataset and an independent validation dataset. The first two periods were discarded to eliminate startup transients associated with rotor acceleration, initialization of the electrical states, and the initial settling of the mechanical motion. Periods 3 and 4 were then averaged to reduce the influence of random measurement noise while preserving the repeatable deterministic response to the periodic excitation signal, thereby providing a clean training dataset (y_{avg}, u_{avg}) .

The averaged response (y_{avg}, u_{avg}) was used as the training dataset for grey-box estimation, whereas the 5th period was reserved exclusively for model validation. This separation ensures that the reported validation fit reflects the predictive capability of the identified model rather than merely its ability to reproduce the estimation data. The identification was performed using a grey-box framework with a fixed-step Runge-Kutta 4 (ODE4) solver with a step size matching the data sampling time, which improved numerical consistency during the estimation process.

3.8 Model Fit on the Training Dataset

The estimated nonlinear grey-box model was first evaluated on the averaged training dataset. As illustrated in Figure 5, the identified model demonstrates a high degree of fidelity when tracking the averaged training data (y_{avg}, u_{avg}) , achieving a fit of 89.78%. The model successfully captures the primary non-linear oscillation characteristics and the phase response induced by the Chirp excitation.

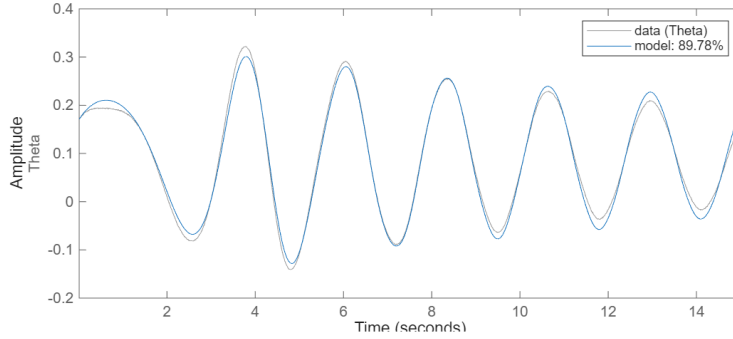


Figure 5: Fit of the identified nonlinear grey-box model to the Chirp-excited training dataset.

The close alignment between the simulated output and experimental data confirms that the identified parameters, including the electrical and thrust-related constants, accurately reflect the dynamic behavior of the helicopter’s pitch axis under active control.

3.9 Model Validation and Results

To assess the generalization capability of the model, a validation study was carried out on the 5th excitation period, which was not used during parameter estimation. The measured pitch-angle response was compared to the simulated output of the identified nonlinear grey-box model under the corresponding measured input voltage. As shown in Figure 6, the identified nonlinear model achieves a fit of **89.69%**.

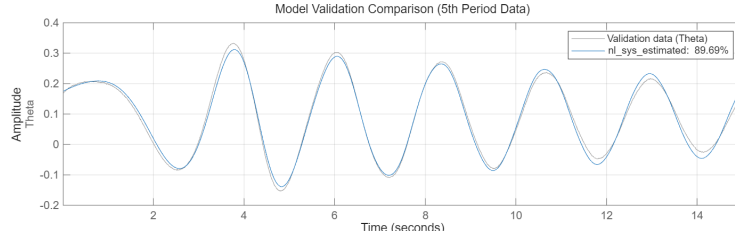


Figure 6: Model validation comparison on the 5th period data, demonstrating a fit of 89.69%.

3.9.1 Discussion

The validation result demonstrates a fit percentage of 89.69%. This indicates that the model successfully captures the dominant coupling between the electrical input and the mechanical response. The slight discrepancies observed near some of the response peaks are likely caused by effects that are not explicitly modeled, such as higher-order aerodynamic nonlinearities, unmodeled friction, variation in rotor efficiency, and high-frequency measurement noise. Nevertheless, the achieved fit is sufficiently high to conclude that the model is well suited for subsequent controller synthesis.

3.10 Identified System Parameters

The complete set of identified parameters are summarized in Table 3.

Table 3: Full System Parameter Identification Results

Category	Parameter	Value	Unit
Mechanical	I_{yy}	0.0399	$kg \cdot m^2$
	m_{heli}	0.7952	kg
	C_{pitch}	0.0039	$N \cdot m \cdot s/rad$
	T_{bias}	-0.0092	$N \cdot m$
Electrical	c	0.0005	$N \cdot m \cdot s^2/rad^2$
	K_m	0.0040	$V \cdot s/rad$
	J	0.0002	$kg \cdot m^2$
	L	0.0148	H
	R	4.1286	Ω
Thrust	C_T	0.3836	$N \cdot s^2/rad^2$

The identified parameter set provides a compact physical representation of the helicopter pitch dynamics. The moment of inertia I_{yy} determines the resistance of the structure to angular acceleration, while the coefficient C_{pitch} characterizes the damping present in the pitch motion. The parameter T_{bias} accounts for constant torque asymmetries caused by mass imbalance or unmodeled static effects. On the actuator side, R and L govern the electrical current dynamics, whereas J and c describe the rotor's rotational inertia and speed-dependent drag. Finally, the thrust coefficient C_T establishes the link between rotor speed and the mechanical torque applied to the pitch axis. Together, these parameters form a physically interpretable model suitable for controller synthesis.

4 Linear Controller Design

The control objective is to achieve accurate tracking of the helicopter pitch angle θ while maintaining closed-loop stability and satisfactory disturbance rejection. Since the identified grey-box model is nonlinear, a local linear approximation is first obtained around the hovering equilibrium point. This linearized model subsequently serves as the common basis for the design of two linear controllers: a Linear Quadratic Gaussian (LQG) Tracker and a Model Predictive Controller (MPC).

4.1 Control Objective and Linearized Plant Model

The identified nonlinear model provides an accurate description of the helicopter pitch dynamics and is well suited for physical interpretation and parameter estimation. However, the synthesis of standard linear control strategies requires a linear time-invariant representation of the plant. For this reason, the nonlinear model is linearized around the hovering operating point, where the helicopter is expected to perform most of its regulation and tracking tasks.

The resulting linear state-space model is written as

$$\dot{x} = Ax + Bu, \quad y = Cx + Du \quad (15)$$

where x is the state vector, u is the motor voltage input, and y is the measured pitch angle. This local approximation captures the dominant small-signal dynamics of the helicopter around hover and forms the basis for both the LQG and MPC controller designs.

4.2 Linearization at Hovering Equilibrium

The hovering equilibrium point (x_0, u_0) was determined by solving the nonlinear equilibrium condition

$$f(x_0, u_0) = 0 \quad (16)$$

subject to the additional requirement that the helicopter is balanced at the nominal reference position $\theta = 0$. Physically, this equilibrium corresponds to the operating condition in which the thrust-generated moment exactly balances gravitational and bias torques, resulting in zero state derivatives.

In practice, the equilibrium point was obtained numerically using nonlinear root-finding procedure. The resulting input u_0 represents the balancing motor voltage required to maintain the helicopter at the hovering operating condition.

After computing the equilibrium state and the corresponding balancing voltage, the nonlinear model was linearized numerically using Jacobian approximation:

$$A = \left. \frac{\partial f}{\partial x} \right|_{(x_0, u_0)}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{(x_0, u_0)} \quad (17)$$

The output matrix was defined as

$$C = [1 \quad 0 \quad 0 \quad 0], \quad D = 0 \quad (18)$$

since only the pitch angle θ is directly measured.

The eigenvalues of matrix A were analyzed to assess the open-loop dynamics of the linearized plant. This analysis provides the foundation for controller synthesis and gives insight into the dominant dynamic modes that must be stabilized and regulated by the feedback design.

4.3 LQG Tracker Synthesis

The first linear controller selected for this assignment is an LQG Tracker. This controller combines optimal state feedback with integral action and a Kalman-based state estimator. The main motivation for this choice is that it provides a systematic way to achieve accurate reference tracking, zero steady-state error, and state reconstruction when not all states are directly measurable.

4.3.1 LQI Regulator Design

To eliminate steady-state tracking error, the linearized plant was augmented with an integral state defined by

$$\dot{x}_i(t) = \int (r(t) - y(t)) dt \quad (19)$$

where $r(t)$ is the reference signal and $y(t)$ is the measured output. The augmented dynamics can therefore be expressed as

$$\begin{bmatrix} \dot{x} \\ \dot{x}_i \end{bmatrix} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix} \begin{bmatrix} x \\ x_i \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r \quad (20)$$

The regulator gains were synthesized using the Linear Quadratic Regulator (LQR) framework applied to the augmented system. The corresponding quadratic cost function is given by

$$J = \int_0^\infty (x_{aug}^T Q_{aug} x_{aug} + u^T R u) dt \quad (21)$$

where Q_{aug} and R were tuned to achieve a balanced performance between response speed and control effort.

The resulting controller gain is written as

$$K_c = [K_x \quad K_i] \quad (22)$$

where K_x is the state-feedback component and K_i is the integral gain. The relatively large weight on the integral state ensures that the controller strongly penalizes accumulated tracking error, thereby improving steady-state accuracy.

4.3.2 LQE State Estimator Design

Because only the pitch angle θ is directly measured, a state estimator is required to reconstruct the unmeasured internal states. For this purpose, a Linear Quadratic Estimator (LQE), equivalent to a continuous-time Kalman filter, was designed for the linearized plant.

The observer dynamics are written as

$$\dot{\hat{x}} = A\hat{x} + Bu + K_f(y - C\hat{x}) \quad (23)$$

where $\hat{x}$ denotes the estimated state vector and K_f is the observer gain matrix. By solving the algebraic Riccati equation K_f was determined, minimizing the estimation error covariance in the presence of sensor noise.

4.3.3 LQG Tracker Implementation

The final LQG Tracker is obtained by combining the LQI regulator and the LQE observer into a single dynamic output-feedback compensator. The controller operates on both the tracking error $e(t) = r(t) - y(t)$ and the measured output $y_m(t)$, resulting in the augmented controller state vector

$$x_K(t) = [\hat{x}(t)^T \quad x_i(t)]^T \quad (24)$$

The synthesis yields the following dynamic feedback law:

$$\dot{x}_K(t) = \begin{bmatrix} A - BK_x - K_f C & -BK_i \\ 0 & 0 \end{bmatrix} x_K(t) + \begin{bmatrix} 0 & K_f \\ 1 & 0 \end{bmatrix} \begin{bmatrix} e(t) \\ y_m(t) \end{bmatrix} \quad (25)$$

where the control law is

$$u(t) = [-K_x \quad -K_i] x_K(t) \quad (26)$$

This implementation allows the controller to simultaneously estimate the system states, regulate the plant, and eliminate steady-state tracking error. As a result, the LQG Tracker provides a structured solution for controlling the pitch axis using only the measured pitch-angle output.

4.3.4 Controller Parameter Tuning and Robustness

The performance of the LQG Tracker is highly dependent on the selection of weighting matrices in the LQI regulator (Q_{aug}, R) and the noise covariance matrices in the LQE estimator (W, V). To ensure optimal tracking and robust estimation, a systematic tuning procedure was implemented.

The LQI weighting matrix Q_{aug} was selected to balance tracking precision and control effort, with specific emphasis placed on the integral error state to eliminate steady-state offsets. For the LQE estimator, the process noise covariance matrix W was configured as a diagonal matrix $W = \text{diag}([1, 10, 0.1, 10^5])$. This non-uniform weighting is physically motivated by the system's dynamics: a significantly higher weight is assigned to the rotor angular velocity (ω) to accommodate its rapid response and the model uncertainty inherent in the linearized electrical-thrust coupling.

This high-gain strategy ensures that the observer dynamics remain faster than the regulator dynamics, as required by the Separation Principle. The measurement noise covariance $V = 10^{-3}$ ensures that the controller remains insensitive to high-frequency sensor noise. Stability of the resulting closed-loop system was verified through pole-zero analysis, confirming that all observer poles are placed deeper in the left-half plane than the regulator poles which is shown in Figure 7, thereby guaranteeing robust state estimation.

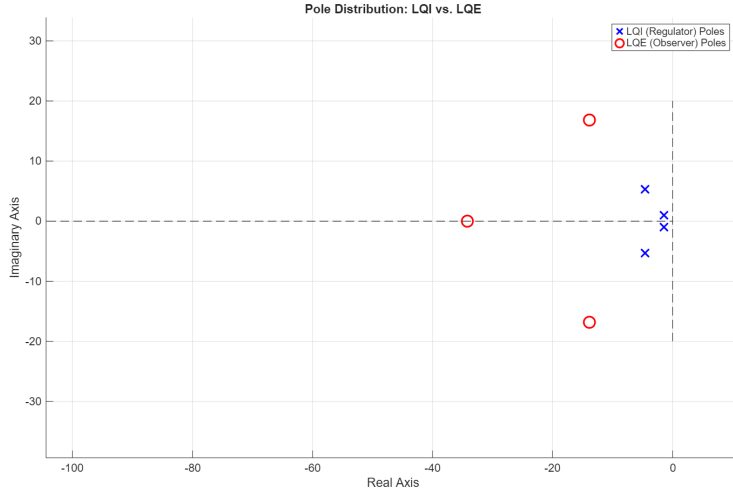


Figure 7: Pole locations associated with the LQI regulator and LQE observer dynamics.

4.3.5 Closed-Loop Stability and Discretization

The final controller was formulated as a dynamic compensator connecting the tracking error and the plant output to the control signal u . The continuous-

transient and without steady-state offset. This confirms the effectiveness of the integral action introduced in the LQI formulation. In addition, the smooth tracking behavior indicates that the state estimator provides sufficiently accurate state reconstruction for the feedback law to operate reliably.

Overall, the simulation results show that the LQG Tracker provides a suitable control solution for pitch-angle regulation of the linearized helicopter model and forms a solid baseline for comparison with the MPC design.

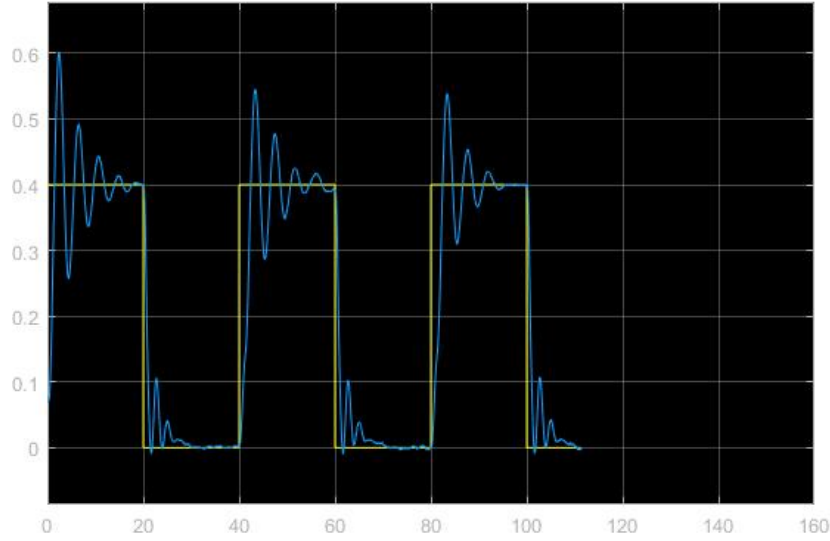


Figure 9: Discrete-time LQG step tracking response of the linearized helicopter model.

4.4 Model Predictive Control (MPC) Design

The second linear controller selected for this assignment is a Model Predictive Controller (MPC). In contrast to the LQG approach, which uses a fixed observer-based feedback structure, MPC determines the control action online by solving a finite-horizon optimization problem at each sampling instant. This makes it particularly suitable for applications in which future system behavior and performance trade-offs must be explicitly accounted for.

4.4.1 Control Principle

The underlying principle of MPC is to predict the future output response of the system over a finite prediction horizon using the linearized discrete-time state-space model. Based on these predictions, the controller computes a sequence of future control inputs that minimizes a predefined cost function while satisfying

the system dynamics. At each sampling instant, only the first element of the optimal control sequence is applied to the plant. At the next time step, the optimization is repeated using updated state information, resulting in the well-known receding-horizon control strategy.

A key advantage of MPC is that it provides a transparent framework for balancing tracking accuracy and control effort. Furthermore, it can be extended naturally to include actuator and state constraints, which makes it highly attractive for practical implementation.

4.4.2 MPC Formulation

The discrete-time MPC controller minimizes the following finite-horizon quadratic cost function:

$$J = \sum_{j=1}^{N_p} \|y(k+j|k) - r(k+j)\|_Q^2 + \sum_{j=0}^{N_c-1} \|\Delta u(k+j|k)\|_R^2 \quad (28)$$

where N_p denotes the prediction horizon, N_c denotes the control horizon, Q is the output tracking weight, and R is the control increment weight. The first term penalizes deviations between the predicted output and the reference trajectory, whereas the second term penalizes aggressive changes in the control input.

The controller was designed using the same linearized model employed for the LQG Tracker. The continuous-time plant was first discretized using a zero-order hold assumption with sampling time $T_s = 0.01$ s. Since the controller is synthesized around the hovering equilibrium point, the MPC formulation operates on the linearized deviation model, such that the optimized control signal corresponds to a voltage variation around the equilibrium input u_0 .

4.4.3 Tuning Parameters

The MPC tuning parameters were selected as follows:

- Prediction Horizon: $N_p = 50$
- Control Horizon: $N_c = 5$
- Output Weight: $Q = 100$
- Input Rate Weight: $R = 0.1$

The chosen prediction horizon allows the optimizer to account for the dominant oscillatory response of the helicopter over a sufficiently long future interval, while the shorter control horizon reduces computational burden and prevents unnecessarily aggressive actuation. The relatively high output weight prioritizes accurate pitch-angle tracking, whereas the input-rate penalty encourages smooth changes in the motor voltage.

In addition, practical constraints were imposed on the manipulated input and its rate of change. These constraints improve realism and prevent the controller from demanding unrealistically large voltage deviations or excessively abrupt actuator commands.

4.4.4 Performance Validation

The discrete-time MPC implementation was validated through numerical simulation using step-reference tracking around the hovering operating point. Since the controller operates on the linearized deviation model, the optimized input corresponds to a voltage variation around the equilibrium input required for hover.

The simulation results demonstrate that the MPC controller is capable of achieving accurate pitch-angle regulation while maintaining smooth control action. As illustrated in Figure 10, the controller exhibits well-damped tracking behavior and satisfactory reference-following capability. Compared to conventional fixed-gain control, the MPC framework provides increased flexibility by explicitly predicting future system behavior and adjusting the control signal accordingly.

These results indicate that MPC forms a suitable alternative to the LQG Tracker for pitch-angle regulation of the 1-DOF helicopter, particularly in applications where predictive action and explicit handling of constraints are advantageous.

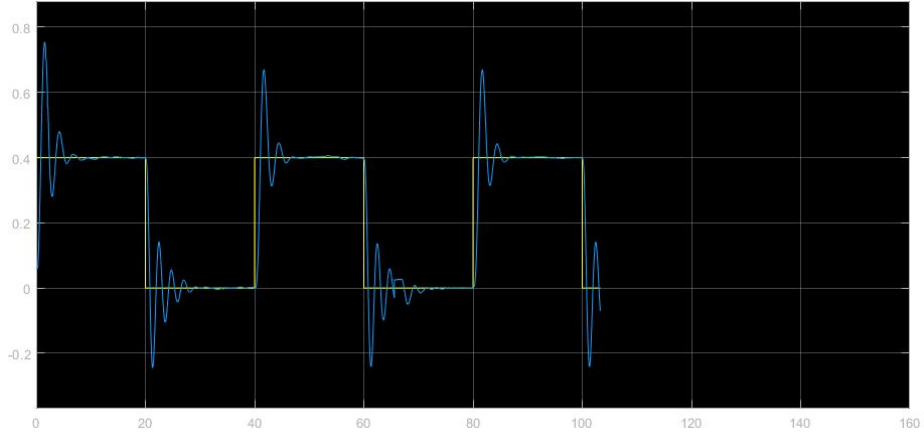


Figure 10: Discrete-time MPC step tracking response of the linearized helicopter model.

4.5 Comparison of the Designed Controllers

Both controllers were designed using the same linearized plant model and both aim to achieve accurate pitch-angle tracking around the hovering operating

point. However, the two approaches differ significantly in structure and design philosophy.

The LQG Tracker combines an optimal regulator with a Kalman-based state estimator and integral action. Its main strength lies in its systematic formulation and relatively low online computational burden after the gain matrices have been computed. This makes it attractive for real-time implementation in systems with limited computational resources. Furthermore, the observer enables the reconstruction of internal states that are not directly measurable.

The MPC controller, on the other hand, determines the control action online by solving a finite-horizon optimization problem at every sampling instant. This yields a more flexible design framework, since future system behavior is explicitly incorporated into the control decision. In addition, MPC provides a natural basis for handling actuator and state constraints, which is an important practical advantage for implementation on real hardware.

From a performance point of view, both controllers are suitable for the pitch-control problem considered in this work, but clear differences can be observed in their transient responses. With the current tuning used in this project, the LQG Tracker shows an overshoot of approximately 30% of the step amplitude and requires about 15 s to settle, as could be seen in Figure 9. This response is relatively more oscillatory, although it remains acceptable and provides reliable reference tracking. The MPC controller on the other hand exhibits a larger overshoot of approximately 60%, but it settles significantly faster, within about 5 s, as could be seen in Figure 10. This indicates that the MPC achieves a more aggressive response, reaching the desired reference more rapidly at the expense of a larger initial peak. It should be noted, however, that these transient characteristics are tuning-dependent; different selections of the LQG weighting matrices or the MPC horizons and cost-function weights could lead to substantially different overshoot and settling behavior.

Overall, the LQG Tracker provides an efficient observer-based solution with moderate overshoot and slower, more oscillatory settling behavior, whereas MPC offers faster convergence and greater flexibility, but with a more pronounced overshoot. Therefore, the final selection between the two strategies depends on the desired trade-off between computational complexity, implementation simplicity, transient speed, and overshoot tolerance.

5 Conclusion

This report presented the modeling, identification, and control design of a 1-DOF helicopter with motion restricted to the pitch axis. A nonlinear grey-box model was first developed by combining the mechanical pitch dynamics with the electrical and rotational dynamics of the actuator. In order to improve parameter identifiability, the model parameters were estimated in two stages: first through a free-swinging pendulum experiment to identify the mechanical parameters, and then through a Chirp-excited experiment to identify the remaining electrical and thrust-related parameters.

The resulting nonlinear model achieved high fidelity with the measured data in both the pendulum and active-excitation experiments, demonstrating that the adopted modeling assumptions and identification procedure provide an accurate representation of the dominant helicopter dynamics. The identified model was subsequently linearized around the hovering equilibrium point and used for the design of two linear controllers: an LQG Tracker and a Model Predictive Controller.

Both control strategies were shown to provide satisfactory pitch-angle tracking performance. The LQG Tracker offers a compact observer-based solution with integral action and low online computational cost, whereas MPC provides greater flexibility through its predictive optimization-based formulation. Overall, the developed model and controllers provide a suitable basis for accurate pitch-angle regulation of the helicopter and for future experimental implementation on the physical setup.

Recommendation

Although the developed controllers meet the design objectives, several avenues for further improvement can be explored to enhance performance and robustness. First, to mitigate the performance asymmetry observed between upward and downward pitch motions, **gain scheduling** can be implemented by designing multiple controllers for different operating regions. Additionally, transitioning to **Nonlinear Model Predictive Control (NMPC)** would allow for the explicit integration of nonlinear system dynamics—such as gravitational restoration and thrust coupling—directly into the optimization framework, providing optimal control across a wider operational envelope. Besides, increasing the sampling frequency is proposed to reduce phase lag induced by zero-order hold discretization, thereby extending the system’s achievable bandwidth and improving tracking speed and transient performance.

6 Appendix: MATLAB Guide

This appendix outlines the steps required to run the MATLAB scripts for system identification and controller design.

6.1 System Identification Procedure

The system identification process consists of two stages:

1. **Pendulum System Identification**

- (a) Run `input_signal_chirp_positive.m`.
- (b) Load the dataset by opening `pendulum_data1.mat`.
- (c) Run `Grey_box_Pendulum.m`.

2. **Complete System Identification**

- (a) Run `input_signal_chirp_positive.m`.
- (b) Load the dataset by opening `Chirp_up.mat`.
- (c) Run `grey_box_chirp.m`.

6.2 LQG Controller Design

To implement the LQG controller, follow these steps:

1. Complete the full system identification procedure described above.
2. Run `LQG_controller.m`.
3. Run `LQG_tracker_simulink.slx` if the experimental setup is connected.

6.3 MPC Controller Design

To implement the MPC controller, follow these steps:

1. Complete the full system identification procedure described above.
2. Run `MPC_controller.m`.
3. Run `MPC_simulink.slx` if the experimental setup is connected.

Artificial Intelligence Statement

I acknowledge the use of Gemini (<https://gemini.google.com/app>) and Deepseek (<https://chat.deepseek.com/>) to support the development of the project by: Refining language, Assisting with debugging, Acting as a brainstorming partner.